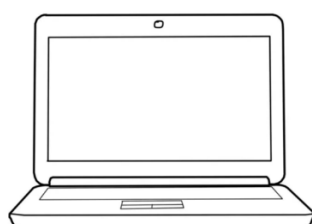


|                                                                                 |                 |                  |
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| Prepared By: AKC                                                                | Checked By: NMR | Approved By: SKP |

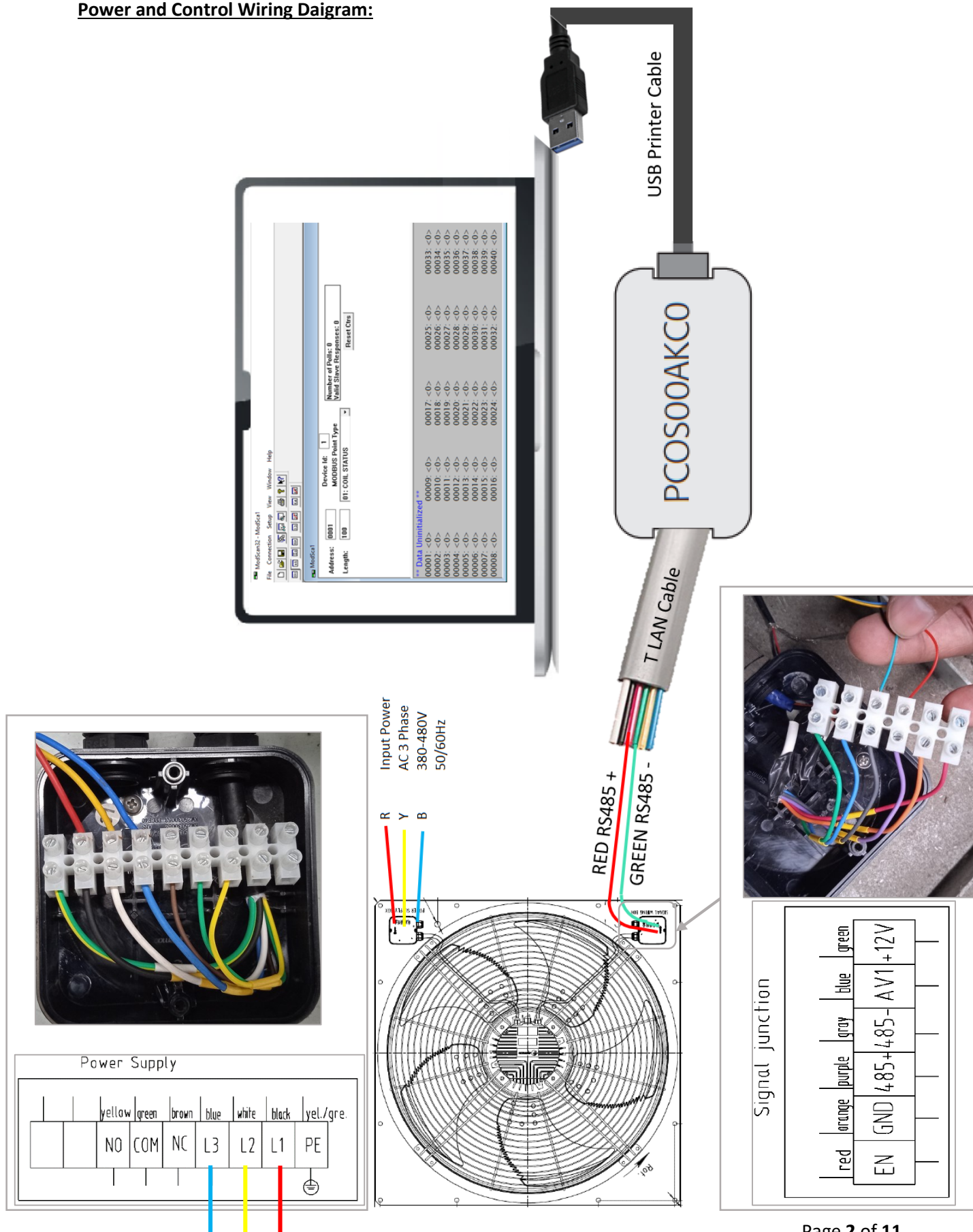
**Scope:** This document will help the user to established a RS485 to USB communication network and configure it's parameter as per KCPL requirement in EC Axial Fan Model No - ODS950C-120B5.DC.V- KCPL manufactured by Kemao and used in Air cooled chillers.



**Required Tools:** EC Axial Fan, Modscan32 software installed Laptop, Carel PCOS00AKC0 Smart Key USB adaptor, Printer USB cable, Telephone-LAN Cable, 380-480V, 50/60Hz, 3 phase input power supply.

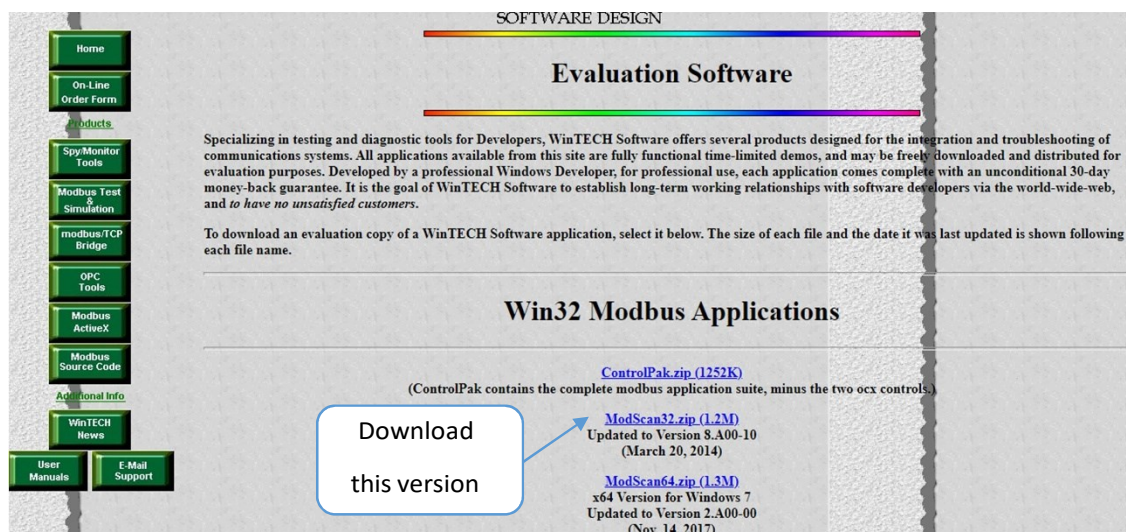


# **Power and Control Wiring Daigram:**



|                                                                                 |                 |                  |
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**MOD Scan Software Installation:** Open any search engine and go to this link--modscan32.zip



Download ModScan32.zip and Extract this file

After extraction the ModScan Zip file have a setup file (Setup file doesn't need to installation in PC). Open setup and ignore all the login section by the cancellation and its ready for testing.

#### **Carel PCOS00AKC0 Smart Key Driver Installation:**

The operation of the adapter requires the installation of the VCP driver, the same used by the USB-RS485 converter, code CVSTDUTLFO or CVSTDUMOR0. Consequently, if one of these devices is already used no new driver needs to be installed. To get the VCP driver (Virtual Com Port), simply go to the CAREL site <http://ksa.carel.com>, under the free download area (USB\_RS-485\_CAREL\_Converter section) or download the file corresponding to the FT232BM component directly from <http://www.ftdichip.com/FTDrivers.htm>. When connecting the adapter for the first time to a USB port on the personal computer, the system detects the presence of new hardware and requests the installation the VCP driver for the management of the serial port; at this point select the path of the directory where the drivers were previously saved and decompressed. If the drivers were installed correctly, the system automatically detects a new COM port, usually COM3. If more than one adapter is connected to the same PC, check under My Computer + right button (mouse) > Properties > Hardware > Peripherals > Ports (COM and LPT) which port has been assigned by the system to the PCOS00AKC0 adapter.

|                                                                                        |                        |                         |
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| <b>Prepared By: AKC</b>                                                                | <b>Checked By: NMR</b> | <b>Approved By: SKP</b> |

#### **Starting Serial Communication with Fan:**

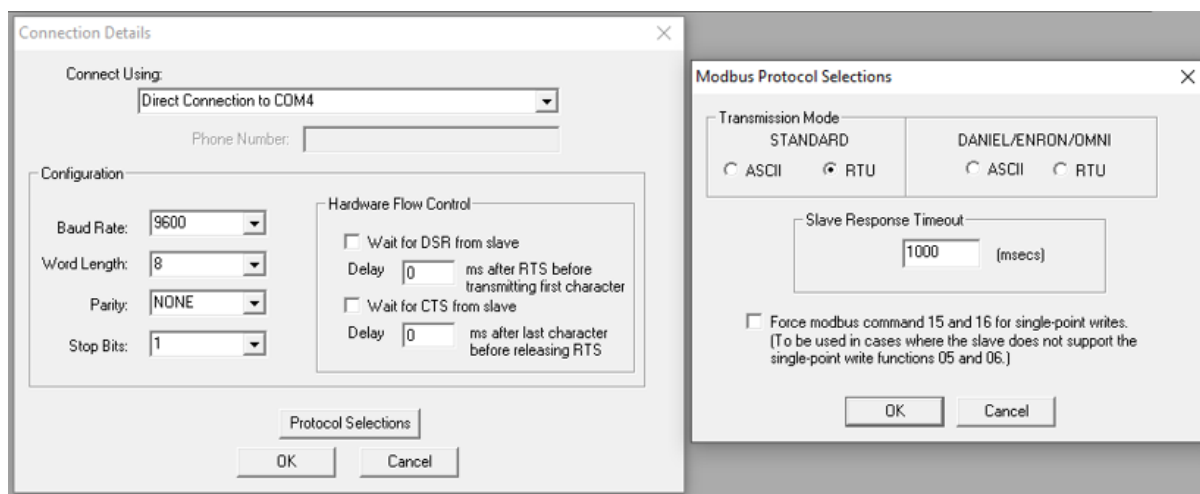
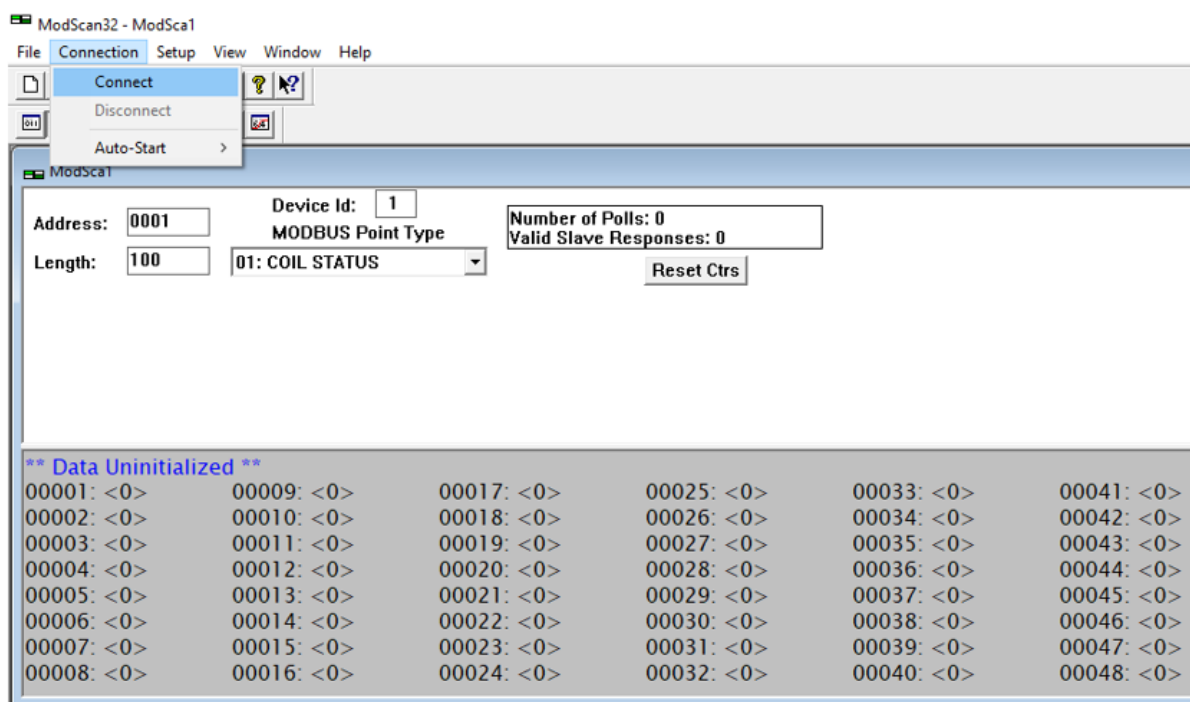
1. Check Power and Control wiring connection as per diagram.
2. Switch on 3 phase input power supply to Fan.
3. Plug in the USB cable to Laptop and observe the communication of Carel PCOS00AKC0 Smart Key USB adaptor with Laptop.
4. Run the ModScan32 Application file in laptop by double click of mouse button.

| Volume (D:) > modscan32_fullversion |                  |                      |        |
|-------------------------------------|------------------|----------------------|--------|
| Name                                | Date modified    | Type                 | Size   |
| Custom1                             | 25-10-1998 10:10 | File                 | 5 KB   |
| Custom2                             | 10-11-1998 21:05 | File                 | 3 KB   |
| ex1.frm                             | 30-10-1997 15:37 | FRM File             | 21 KB  |
| example1                            | 09-05-1997 22:02 | Microsoft Excel C... | 3 KB   |
| modbus1.mdb                         | 05-11-1997 15:53 | MDB File             | 264 KB |
| modbusm.dll                         | 16-02-1999 10:30 | Application exten... | 41 KB  |
| ModScan32.cnt                       | 25-10-1998 12:07 | CNT File             | 2 KB   |
| ModScan32                           | 08-08-2009 23:23 | Application          | 660 KB |
| MODSCAN32                           | 19-09-1999 10:47 | Help file            | 252 KB |
| ModScan32.tlb                       | 19-09-1999 10:47 | TLB File             | 3 KB   |
| ModScan32Ex.vbp                     | 30-10-1997 15:37 | VBP File             | 2 KB   |
| ms32frm.cfg                         | 21-10-2009 17:17 | CFG File             | 1 KB   |
| oilp                                | 09-11-1998 21:45 | BMP File             | 46 KB  |
| otemp                               | 09-11-1998 21:53 | BMP File             | 49 KB  |
| ScanGuide                           | 19-09-1999 10:58 | Text Document        | 32 KB  |
| speedo                              | 09-11-1998 21:41 | BMP File             | 112 KB |
| tach                                | 09-11-1998 21:37 | BMP File             | 76 KB  |
| temp                                | 09-11-1998 21:52 | BMP File             | 49 KB  |
| volt                                | 09-11-1998 21:57 | BMP File             | 49 KB  |
| WS_FTP                              | 24-06-1999 18:56 | Text Document        | 1 KB   |



|                                                                                        |                        |                         |
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| <b>Prepared By: AKC</b>                                                                | <b>Checked By: NMR</b> | <b>Approved By: SKP</b> |

5. Click on Connect and confirm the the connection detail as per below screen short.



We receive following default communication setting which is done by Fan manufacturer.

Baud Rate : 9600

Word Length : 8

Parity : None

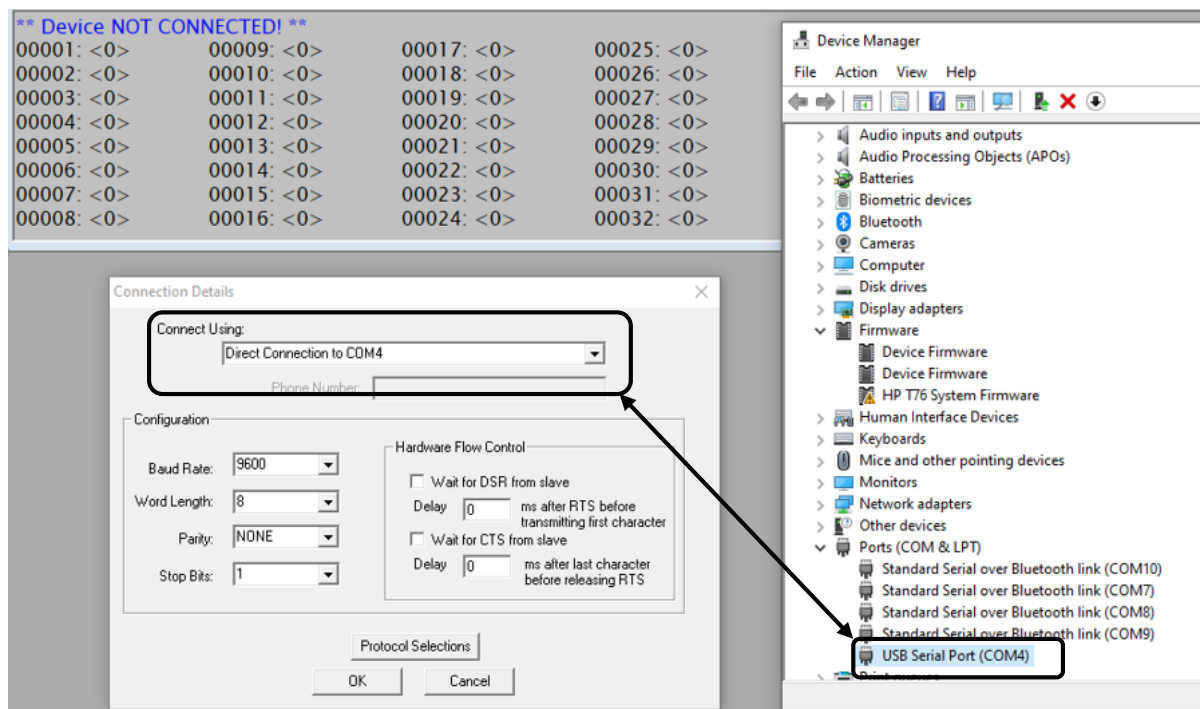
Stop Bit : 1

Modbus Device ID : 1

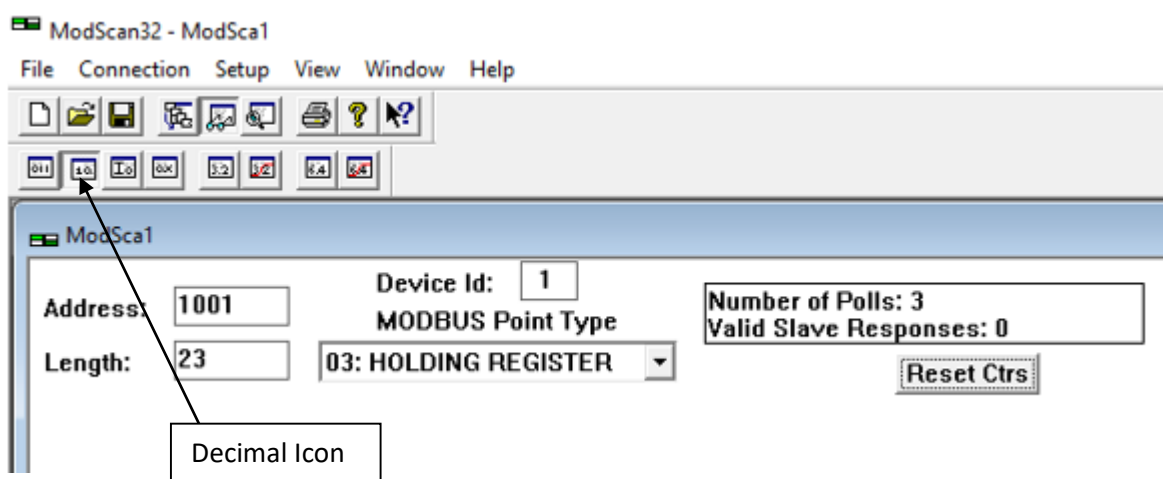
Note - During first time communication at KCPL factory this setting works fine, for next time and onwards Baud rate need to change to 192000.

|                                                                                        |                        |                         |
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- Check the COM port selection in drop down box under heading “Connect using” in this case COM4 USB port is assigned to ModScan32 software for Modbus communication with Fan. Carel PCOS00AKC0 Smart Key USB adaptor must be plugged in this USB port only.

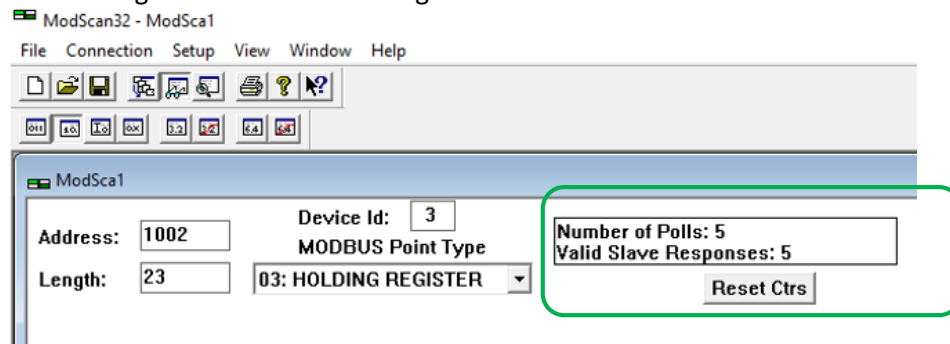


- In ModScan Software screen click on **Decimal** icon tab and fill the input data box as per below image.



|                                                                                 |                 |                  |
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8. Serial communication with fan will be start, see the increasing counter of Valid Slave Response marked in green box of below image.



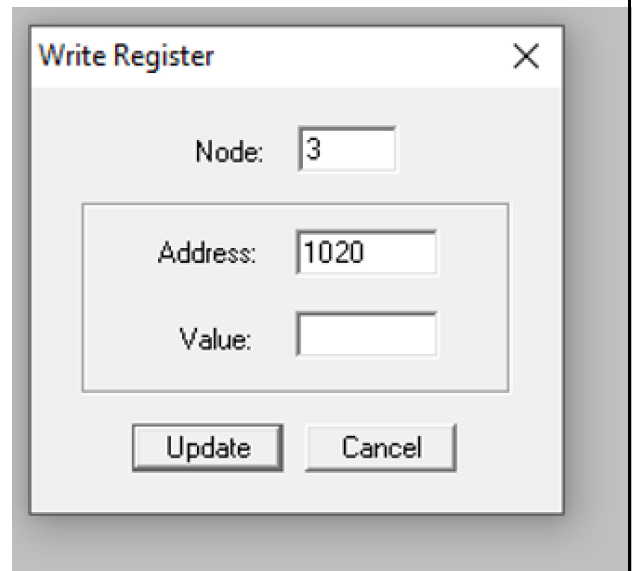
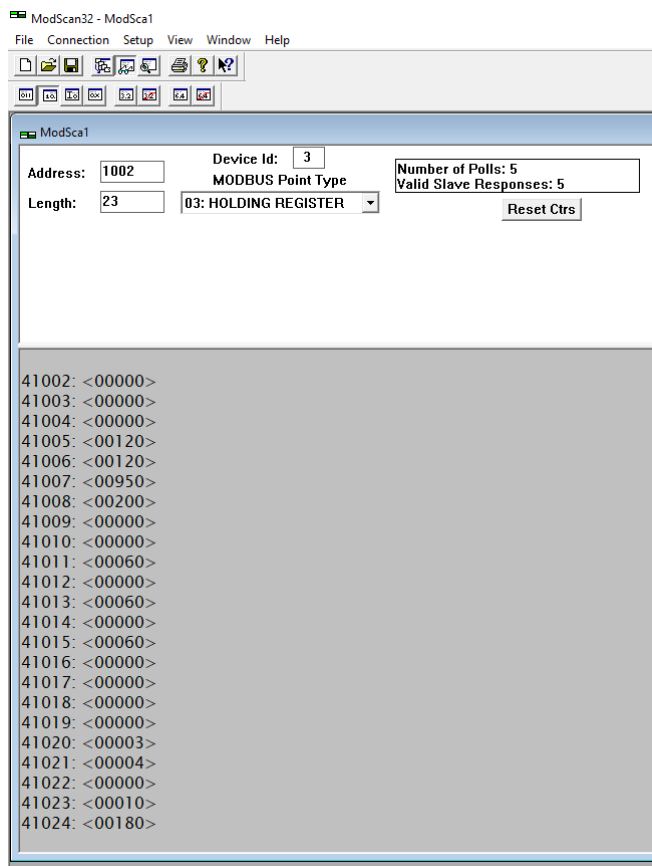
9. Double click to change the Modbus Holding Register 41020 from default value 1 to any suitable number (1 to 20) as per installation location of the fan specified in last page. In this step, we have assign a Device Id (Modbus address number to the connected fan).

## Caution !



Re-confirm and note down the type value editing for Modbus Holding Register 41020 and (take a screenshot also) before click on Update button.

This number has significant role in re-establishing a communication with the fan. Failed to do so Modbus communication will be lost and not possible to re-connect. Defective fan module need to send back to its manufacturer for reset.



|                                                                                 |                 |                  |
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10. Disconnect the input power supply for at least 180 second and then re-connect.  
Write value for the remaining register as per below and verify it.

The screenshot shows the ModScan32 software interface. The top menu bar includes File, Connection, Setup, View, Window, and Help. Below the menu is a toolbar with various icons. The main window is titled 'ModScan32 - ModSca1' and contains the following fields:

- Address: 1002
- Device Id: 20
- MODBUS Point Type: 03: HOLDING REGISTER
- Length: 23
- Number of Polls: 187
- Valid Slave Responses: 187
- Reset Ctrs button

Below these fields is a list of registers with their descriptions:

```

41002: <00000> Start/Stop switch : 0 OFF, 1 ON
41003: <00000> Fan speed (rpm) absolute value
41004: <00000> Direction : 0 Forward, 1 Reverse
41005: <00120> Rising rate (rpm/s)
41006: <00120> Falling rate (rpm/s)
41007: <00980> Speed max (rpm) Can not change
41008: <00200> Speed min (rpm) absolute value
41009: <00000> Communication timeout switch
41010: <00000>
41011: <00060>
41012: <00000>
41013: <00060>
41014: <00000>
41015: <00060>
41016: <00000> Communication timeout switch Bit0:1=0 OFF Bit0:1=2 ON Bit8=0 Empty effective Bit8=1 Grounding effective
41017: <00000>
41018: <00000>
41019: <00000> Clear fault count Bit2 = 1 Clear
41020: <00020> MODBUS address number assigned to the Fan
41021: <00004> Baud rate : 0 - 9600, 1- 1200, 2-2400, 3-4800, 4-19200, 5-38400
41022: <00000> Parity bit Bit1:2=0 NONE Bit1:2=1 ODD Bit1:2=2 EVEN Stop bit Bit3:4=0 1 STOP BIT Bit3:4=1 2 STOP BITS
41023: <00010> Communication reply delay time (ms)
41024: <00180> Communication timeout time (s)

```

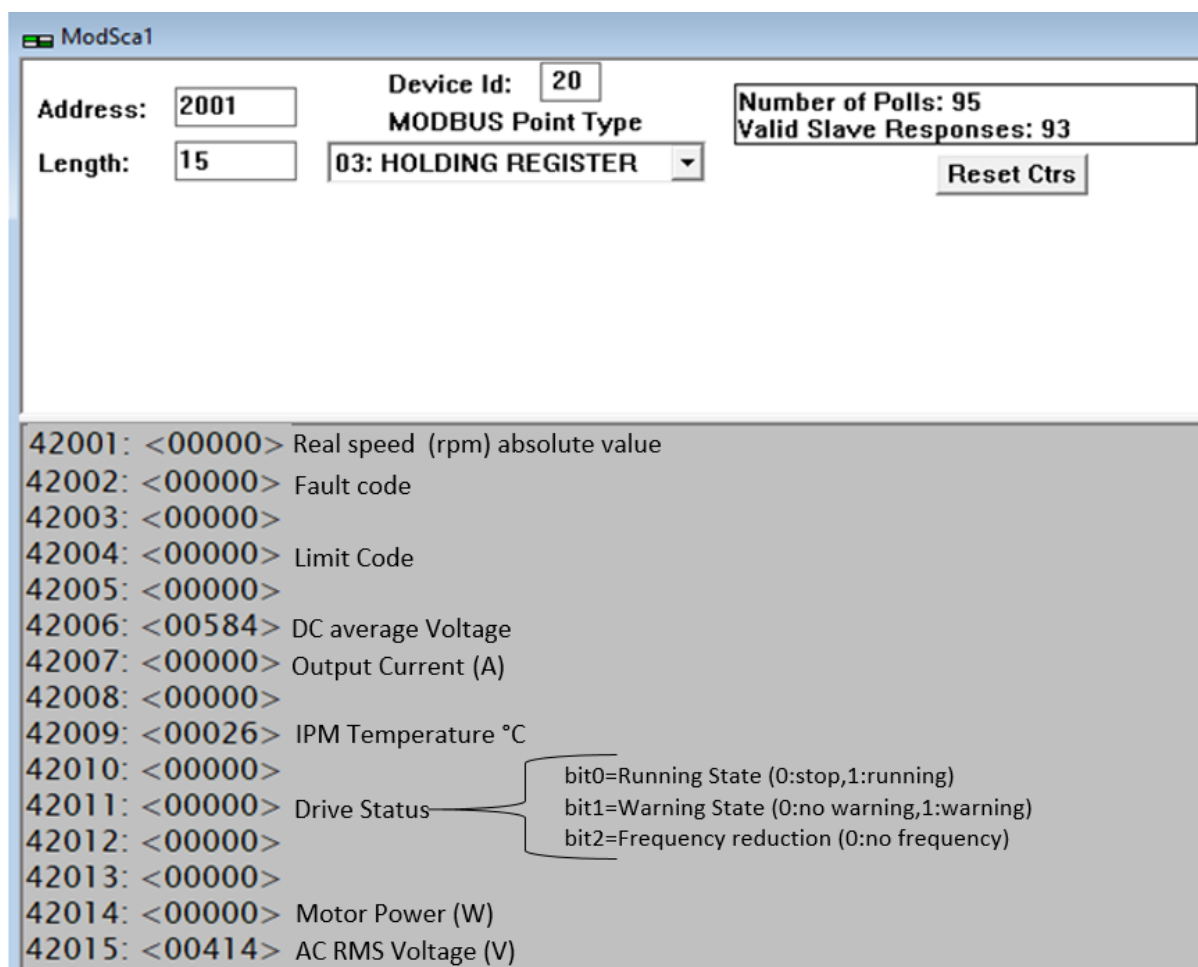
*Note - If the outlet air temperature of the condenser coil is more than 60°C, keep Maximum speed setting to 950 rpm to avoid un-necessary tripping of the fan.*

11. Disconnect the power supply for at least 180 second and re-connect. Stablish communication with ModScan32 software with Baud rate 19200 and check all set point value written in previous stage. There must not be any deviation. If so, repeat the above process.
12. Now, Fan is ready for rotation check test. Write the Register Address 41003 value to 500, then write the Register Address 41002 value to 1. Allow the fan to run for sufficient time and observe for any abnormal noise or any other defect.



|                                                                                 |                 |                  |
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13. Open the Monitoring screen to read the important parameters.



ModSca1

Address:  Device Id:  Number of Polls: 95  
Length:  MODBUS Point Type:  Valid Slave Responses: 93

42001: <00000> Real speed (rpm) absolute value  
42002: <00000> Fault code  
42003: <00000> Limit Code  
42004: <00000> DC average Voltage  
42005: <00584> Output Current (A)  
42006: <00000> IPM Temperature °C  
42007: <00000> Drive Status  
42008: <00000> Motor Power (W)  
42009: <00000> AC RMS Voltage (V)  
42010: <00000> bit0=Running State (0:stop,1:running)  
42011: <00000> bit1=Warning State (0:no warning,1:warning)  
42012: <00000> bit2=Frequency reduction (0:no frequency)  
42013: <00000>  
42014: <00000>  
42015: <00414>

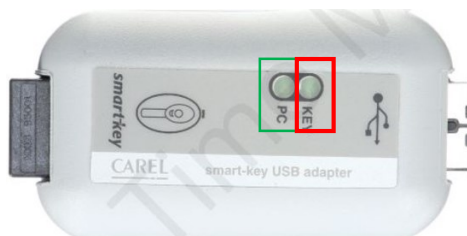
14. If Fan is performing well then to stop it, first change the register address 41002 value to 0 then also change the register address 41003 value to 0.

15. Configuration process is completed now. Disconnect the input power supply, remove the communication wire. It can be use in assembly of Chiller product.

|                                                                                 |                 |                  |
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## Troubleshooting for communication problem

1. If you do not see the increase number in Valid Slave Response counter, check Number of Polls counter and Smart key LED status.



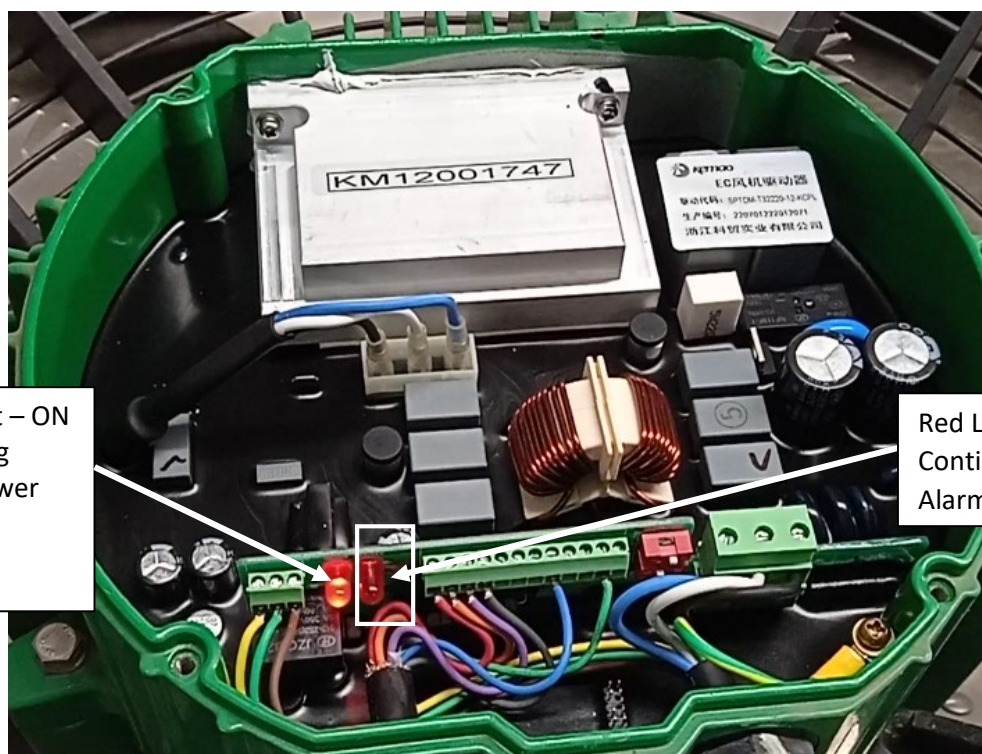
If PC light (marked in Green box) is blinking and Key Light (marked in Red box) is continuously OFF, this is indication of something is wrong in between Smart key and Fan.

Re-check input power supply status to the Fan.

Try with a different Baud rate which is 192000, If still no response is received try to communicate with Fan with a different Modbus Device ID (1 to 20).

2. At last, disconnect input power supply and open the end cover of Fan assembly to check the status of power circuit board.

Once cover plate is open, switch on the power supply and monitor the both LED status

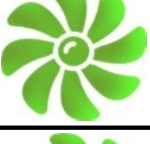
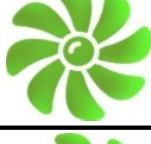
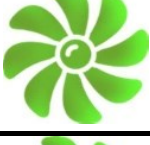


Red Light – ON  
Indicating  
Input Power  
supply is  
Healthy

Red Light - Blinking  
Continuously means No any  
Alarm

3. If input Power supply is OK and there is no any alarm, then re-check control wiring  
Take support from Fan Manufacturer for further troubleshooting.

|                                                                                 |                 |                  |
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| Electric Panel                   |                                                                                     |                                                                                     |                                   |
|----------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------|
| Fan No - 2<br>Modbus address -2  |    |    | Fan No - 1<br>Modbus address -1   |
| Fan No - 3<br>Modbus address -6  |    |    | Fan No - 4<br>Modbus address -5   |
| Fan No - 5<br>Modbus address -10 |    |    | Fan No - 6<br>Modbus address -9   |
| Fan No - 7<br>Modbus address -14 |   |   | Fan No - 8<br>Modbus address -13  |
| Fan No - 9<br>Modbus address -18 |  |  | Fan No - 10<br>Modbus address -17 |
| Fan No - 2<br>Modbus address -20 | <b>Not applicble for 18<br/>Fan chiller model</b>                                   |                                                                                     | Fan No - 1<br>Modbus address -19  |
| Fan No - 3<br>Modbus address -16 |  |  | Fan No - 4<br>Modbus address - 15 |
| Fan No - 5<br>Modbus address -12 |  |  | Fan No - 6<br>Modbus address -11  |
| Fan No - 7<br>Modbus address -8  |  |  | Fan No - 8<br>Modbus address -7   |
| Fan No - 9<br>Modbus address - 4 |  |  | Fan No - 10<br>Modbus address -3  |